

W2-PM1

WEEK 2 | PROJECT MODULE 1



 Cybersecurity

 Ethical Hacking

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FOOTPRINTING & RECONNAISSANCE ATTACKS WITH MULTIPLE KALI TOOLS

CYBERSECURITY & ETHICAL HACKING PROJECT TASKS

WEEK 2 | PROJECT MODULE 1

FOOTPRINTING & RECONNAISSANCE ATTACKS WITH MULTIPLE KALI TOOLS

Background

Reconnaissance (also called as footprinting) is the first step in any real attack or security test. Before touching a target, an attacker quietly collects as much public information about it as possible. This includes who owns the domain, its real IP address, the hosting provider, the web technologies it runs, its DNS and mail records, and whether a firewall is protecting it. All of this comes from information the target has already made public, so the target never even knows it is being studied. This is why recon is powerful and very hard to detect.

What we will do in this task?

In this lab you will footprint the live website networkwalks.com using six built-in Kali Linux tools: whois, whatweb, nslookup, curl, wafw00f and dnsrecon. Each tool reveals a different piece of the target, and together they build a full profile of it. The information you gather here is the foundation for everything that follows. In the next projects/tasks you will use these same findings to plan your scanning and your attacks, because you cannot attack what you have not first understood. Record every output carefully, as you will need it for your final report.

TASKS

Task 1. Run whois to find the domain registration details.



Task 2. Run whatweb to fingerprint the web technologies.

Task 3. Run nslookup to resolve the domain to its IP address.



Task 4. Run curl -I to read the HTTP response headers.

Task 5. Run wafw00f to detect a Web Application Firewall.



Task 6. Run dnsrecon to enumerate all DNS records.



TASK 1

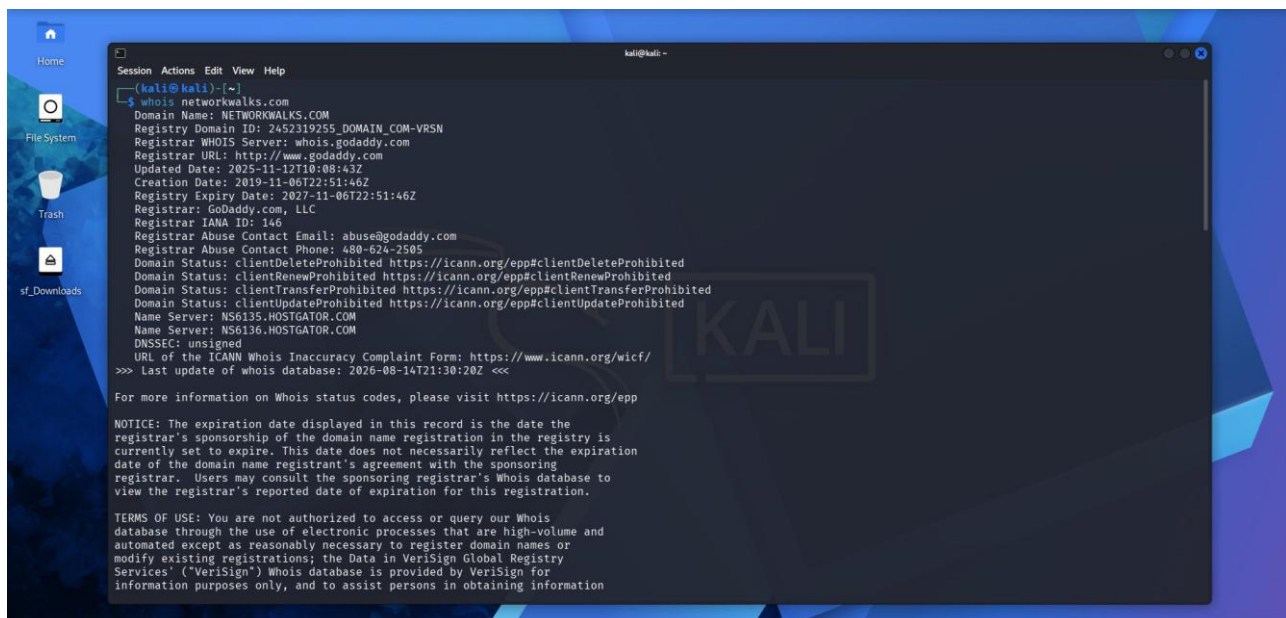
Query the public domain registration record to find who owns the domain, when it was registered, and its name servers.

SOLUTION

Open a terminal in Kali Linux & run this command:



```
$ whois networkwalks.com
```



```

kali@kali ~
$ whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T10:08:43Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-08-14T21:30:20Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information

```

Save the screenshot.

Save the output in a text file.

HOW ATTACKERS USE THIS

whois reveals the registrar, registration and expiry dates, and name servers. Here the name servers point to HostGator, so an attacker instantly learns the hosting provider. Registration dates and abuse contacts help with social engineering and planning.

TASK 2

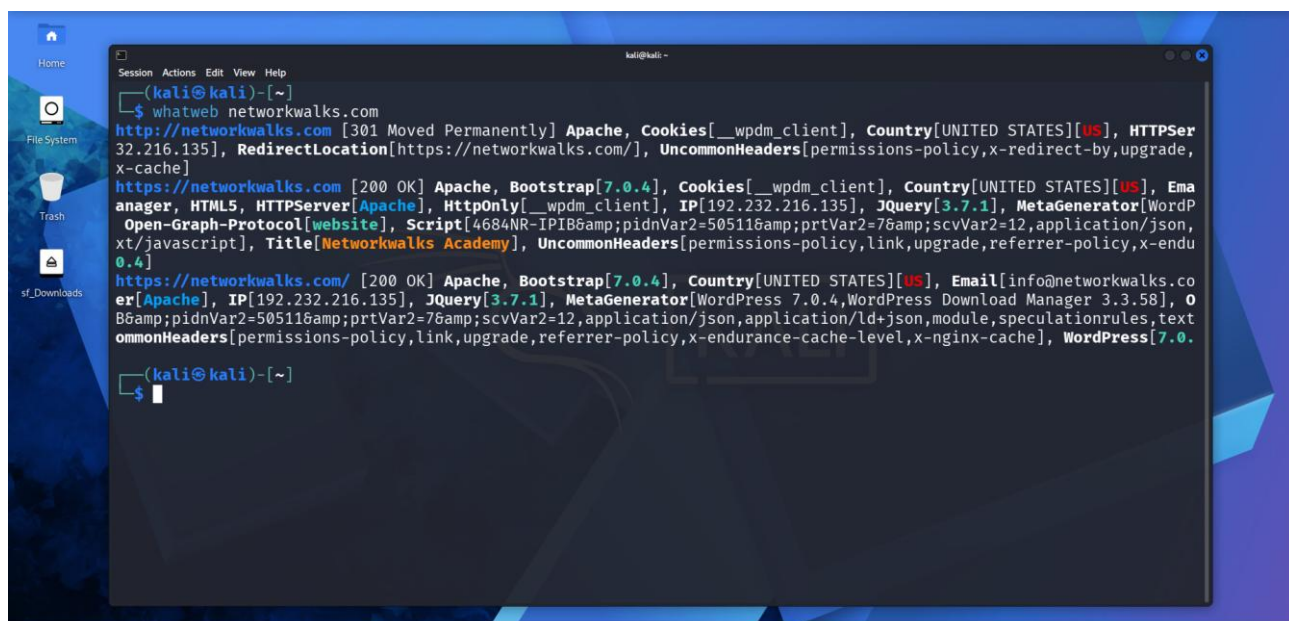
Fingerprint the technologies running on the website: web server, CMS, plugins, frameworks and IP address.



SOLUTION

Open a terminal in Kali Linux & run this command:

```
$ whatweb networkwalks.com
```



```
(kali@kali)-[~]
$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][US], HTTPServer[32.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,x-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.0.4], Cookies[__wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.0.4], Open-Graph-Protocol[website], Script[4684NR-IPID&pidnVar=50511&prtVar=7&scvVar=12,application/json,xt/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.0.4]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.0.4], Country[UNITED STATES][US], Email[info@networkwalks.com], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.0.4,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPID&pidnVar=50511&prtVar=7&scvVar=12,application/json,application/ld+json,module,speculationrules,text/css], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.0.4]
```

Save the screenshot.

Save the output in a text file.

HOW ATTACKERS USE THIS

whatweb exposes the exact software and versions (here Wordpress 7.0.4 and WP Download Manager 3.3.58). An attacker looks these versions up in vulnerability databases to find known exploits. It also leaks the server IP and an email address.

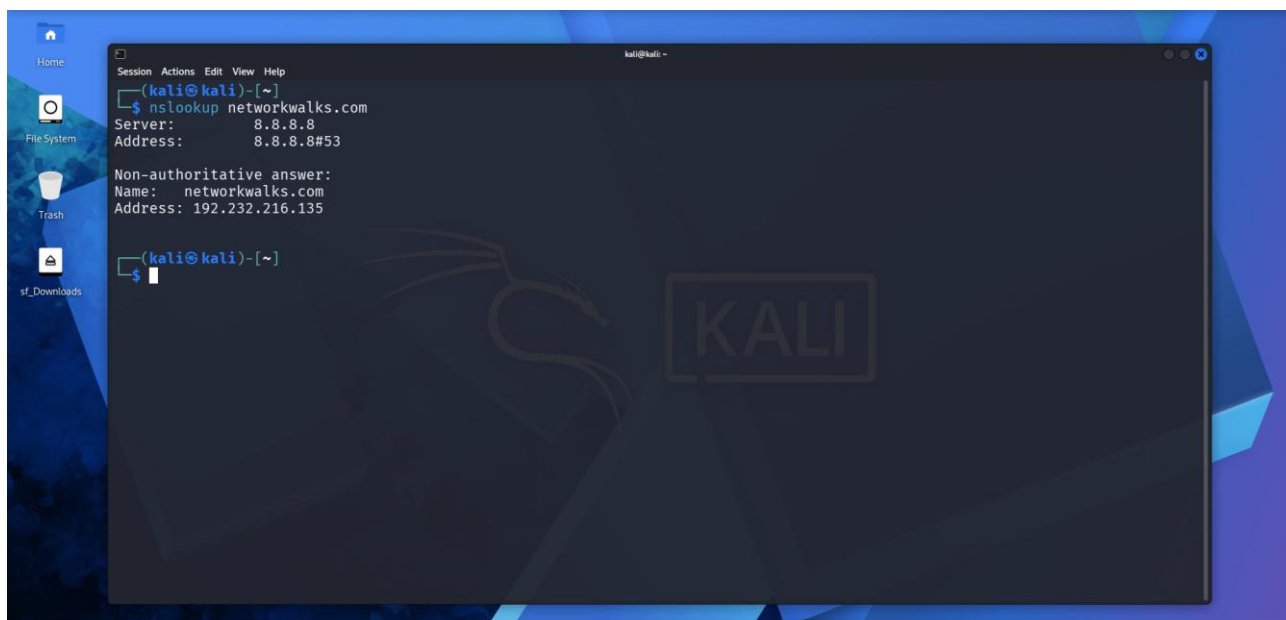
TASK 3

Resolve the domain name to its IP address using DNS.

SOLUTION

Open a terminal in Kali Linux & run this command:

```
$ nslookup networkwalks.com
```



Save the screenshot.

Save the output in a text file.

HOW ATTACKERS USE THIS

nslookup turns a domain name into its real IP address (192.232.216.135). Knowing the IP lets an attacker scan the server directly, look up other sites on the same IP, and map the target's infrastructure.

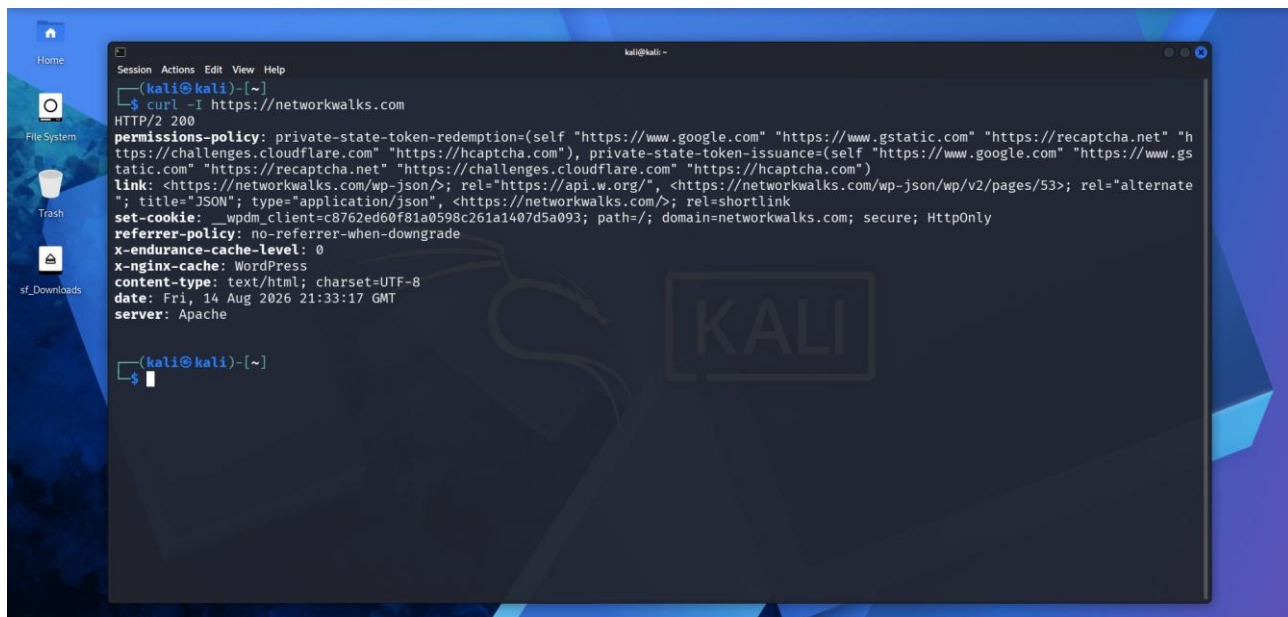
TASK 4

Read the HTTP response headers to see the server banner, status, cookies and redirects.

SOLUTION

Open a terminal in Kali Linux & run this command:

```
$ curl -I https://networkwalks.com
```



```
kali@kali: ~  
$ curl -I https://networkwalks.com  
HTTP/2 200  
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")  
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink  
set-cookie: __wpdm_client=c8762ed60f81a0598c261a1407d5a093; path=/; domain=networkwalks.com; secure; HttpOnly  
referrer-policy: no-referrer-when-downgrade  
x-endurance-cache-level: 0  
x-nginx-cache: WordPress  
content-type: text/html; charset=UTF-8  
date: Fri, 14 Aug 2026 21:33:17 GMT  
server: Apache  
  
kali@kali: ~  
$
```

Save the screenshot.

Save the output in a text file.

HOW ATTACKERS USE THIS

HTTP headers leak the web server, caching stack and hidden endpoints (here the WordPress REST API at /wp-json/). Attackers read headers to fingerprint the stack and find entry points without even loading the full page.

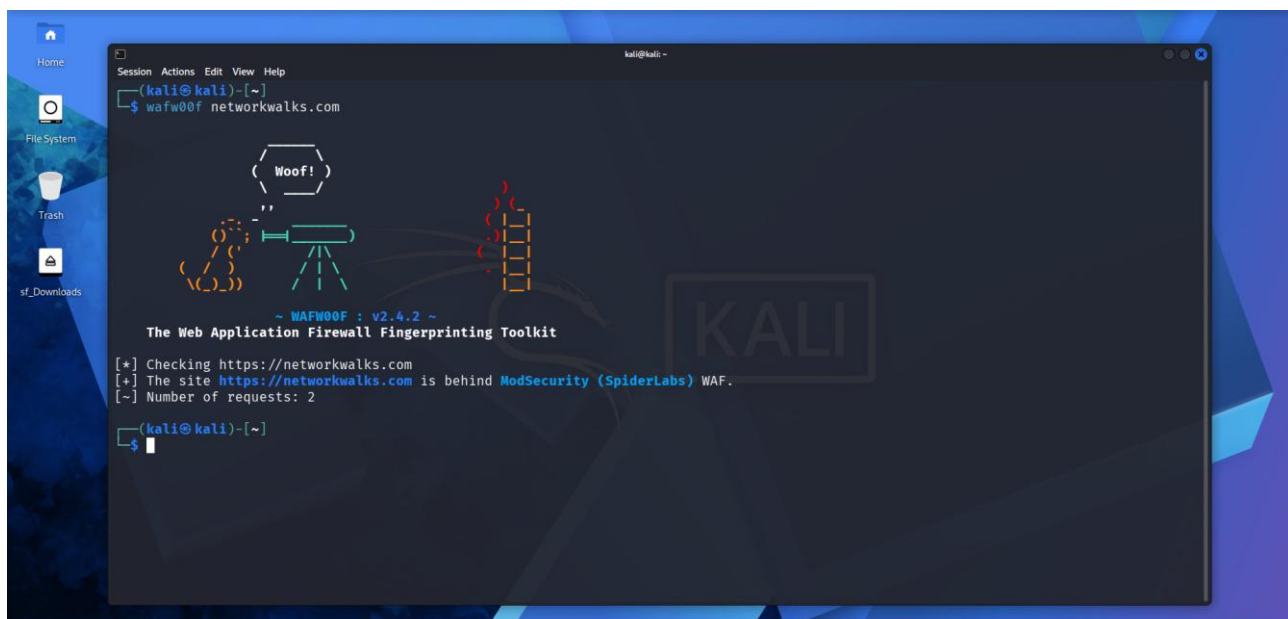
TASK 5

Detect whether a Web Application Firewall (WAF) is protecting the target site.

SOLUTION

Open a terminal in Kali Linux & run this command:

```
$ wafw00f networkwalks.com
```



Save the screenshot.

Save the output in a text file.

HOW ATTACKERS USE THIS

wafw00f tells an attacker if a firewall is watching. Here the site sits behind ModSecurity (SpiderLabs). Knowing a WAF is present shapes the whole attack: naive attempts will be blocked or logged, so the attacker must adapt or try to bypass it.

TASK 6

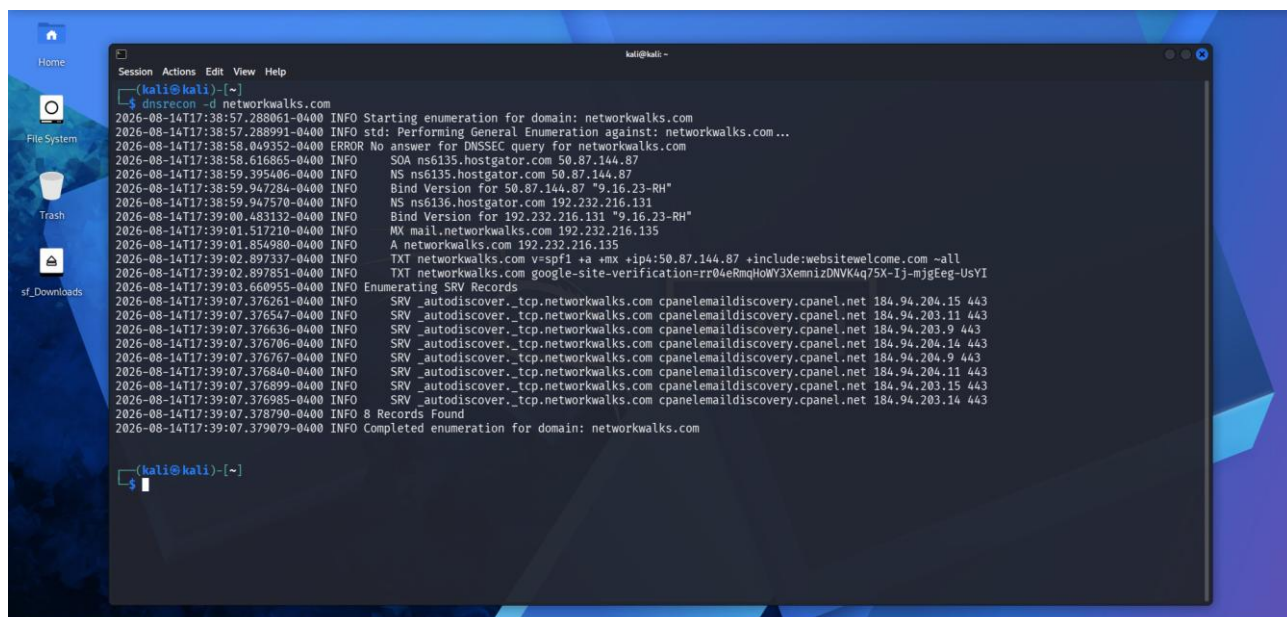
Enumerate all DNS records: name servers, mail servers, SPF, TXT and service (SRV) records.

SOLUTION



Open a terminal in Kali Linux & run this command:

```
$ dnsrecon -d networkwalks.com
```



```
(kali@kali)~$ dnsrecon -d networkwalks.com
2026-08-14T17:38:57.288061-0400 INFO Starting enumeration for domain: networkwalks.com
2026-08-14T17:38:57.288991-0400 INFO std: Performing General Enumeration against: networkwalks.com...
2026-08-14T17:38:58.049352-0400 ERROR No answer for DNSSEC query for networkwalks.com
2026-08-14T17:38:58.616865-0400 INFO SOA ns6135.hostgator.com 50.87.144.87
2026-08-14T17:38:59.395406-0400 INFO NS ns6135.hostgator.com 50.87.144.87
2026-08-14T17:38:59.947284-0400 INFO Bind Version for 50.87.144.87 "9.16.23-RH"
2026-08-14T17:38:59.947570-0400 INFO NS ns6136.hostgator.com 192.232.216.131
2026-08-14T17:39:00.483132-0400 INFO Bind Version for 192.232.216.131 "9.16.23-RH"
2026-08-14T17:39:01.517210-0400 INFO MX mail.networkwalks.com 192.232.216.135
2026-08-14T17:39:01.854980-0400 INFO A networkwalks.com 192.232.216.135
2026-08-14T17:39:02.897337-0400 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com -all
2026-08-14T17:39:02.897851-0400 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemniZDNVKAq75X-lj-mjgEg-UsYI
2026-08-14T17:39:03.660955-0400 INFO Enumerating SRV Records
2026-08-14T17:39:07.376261-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.15 443
2026-08-14T17:39:07.376547-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.11 443
2026-08-14T17:39:07.376636-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.9 443
2026-08-14T17:39:07.378706-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.14 443
2026-08-14T17:39:07.378767-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.9 443
2026-08-14T17:39:07.378840-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.11 443
2026-08-14T17:39:07.378899-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.15 443
2026-08-14T17:39:07.378985-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.14 443
2026-08-14T17:39:07.378790-0400 INFO 8 Records Found
2026-08-14T17:39:07.379079-0400 INFO Completed enumeration for domain: networkwalks.com
(kali@kali)~$
```

Save the screenshot.

Save the output in a text file.

HOW ATTACKERS USE THIS

dnsrecon maps the target's entire DNS footprint: mail servers, DNS software version (Bind 9.16.23), SPF policy and cPanel service records. Each record is a potential foothold and helps an attacker understand the email and hosting setup.

-End-

WHY FOOTPRINTING MATTERS?

Reconnaissance is the first stage of every real attack. Before touching a target, an attacker quietly builds a complete profile of it using only public information, exactly the tools in this task. whois and DNS tools (nslookup, dnsrecon) reveal who owns the domain, its real IP address, its hosting provider and its mail servers. whatweb and curl fingerprint the exact software and versions running, which an attacker matches against known vulnerabilities. wafw00f warns them whether a firewall is watching, so they know how careful to be.

None of these tools attack the target. They only read what is already public, which is exactly why footprinting is so powerful and so hard to detect. The more an organization leaks, the easier every later stage of the attack becomes. This is also why defenders run the same tools on themselves: to see what an attacker would see, and to reduce it.

Extra References & tips



Reconnaissance (footprinting) is the very first phase of ethical hacking and real attacks.



Passive recon only reads public information and never touches the target, making it hard to detect.



Kali Linux ships with dozens of recon tools built in — whois, whatweb, dnsrecon and more. You can try more tools after this lab.



The less information an organization leaks publicly, the harder every later attack stage becomes.



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Virtual lab

Your own devices



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